

Particulate Matter Monitoring & Health Effects

Issue 2026-10-06 | Entry window 6 October 2026 (full day) | 11 records screened in scope

11RECORDS IN SCOPE
(61 EXCLUDED)**6**OF 10 SUBTOPIC
BANDS POPULATED**0**EFFECT ESTIMATES
WITH A CI**2**TIER-A
RECORDS**4**RECORDS CARRIED ON
METADATA ALONE

Signal of the day

An emission-control policy moved mean particle diameter by 27 % — which is a calibration drift no sensor firmware will ever report.

Dai and colleagues (*ACP*) ran single-particle aerosol mass spectrometry through winter and summer emission-control periods in Yangzhou against matched normal periods, defining sulfur-to-nitrogen ratios in the gas phase, the particle phase and by particle number. Cutting NO_x while SO_2 stayed flat raised the gas-phase ratio, lifted the particle-phase ratio by $\sim 57\%$ across particle classes, and expanded mean vacuum aerodynamic diameter by 27 %. Enrichment was largest in particles carrying both black and organic carbon, least in BC-free inorganic particles. **Caveat:** control periods are confounded with season and meteorology, and “causal inference” is a strong label for an observational series with co-varying controls. **Why it matters for sensing:** a nephelometric node’s mass estimate is a fixed assumption about size distribution and refractive index wrapped around a scattering measurement. A 27 % shift in mean diameter changes the scattering cross-section per unit mass directly — so a network calibrated before an emission control reads differently after it, with no hardware change and nothing in the diagnostics to show for it. Co-location drift studies attribute this to the sensor. It is not the sensor.

What changed today

- **Infiltration above 0.6 in naturally ventilated classrooms.** Ten Klang Valley primary schools across five land-use classes: indoor 24 h $\text{PM}_{2.5}$ peaked at mixed industrial/port schools ($35.33 \pm 9.25 \mu\text{g m}^{-3}$), outdoor at high-traffic schools (34.50 ± 20.64), with mean $F_{\text{inf}} > 0.6$. Two buildings per class confounds land use with building, and F_{inf} from indoor-outdoor ratios absorbs any indoor source — but if it holds, outdoor network data is a usable proxy for children’s exposure in this climate.
- **Brain structure, without numbers.** 4,276 adults with repeated MRI: higher long-term $\text{PM}_{2.5}$ and PM_{10} associated with accelerated grey-matter loss and greater white-matter-hyperintensity progression. The abstract reports *no* effect size, interval, exposure contrast, cohort name or country, so the record is logged and cannot be entered as an estimate.
- **Italy’s national attributable-burden platform.** $\text{PM}_{2.5}$ and NO_2 modelled at 1 km by random forest for 2016–2024, population-weighted to municipality level against the 2021 census, and run through HRAPIE-2 functions to the WHO $5 \mu\text{g m}^{-3}$ counterfactual. The surface is validated against the monitors that train it, and municipality-scale population weighting erases the within-city contrast — which is precisely the gap a dense network fills.
- **Port outflow at $4\text{--}10 \mu\text{g m}^{-3}$.** Four seasons of HR-ToF-AMS at a coastal receptor downwind of Busan: five PMF organic factors, frequent ultrafine growth events, and oxidised primary organic aerosol reaching 36 % in autumn under north-easterly port transport. Growth events at that mass loading are invisible to a mass-reporting network.

Window continuity

The 5 October issue closed with `last_entry_date` at 5 October. This issue collects **6 October in full**, continuous with it, and is the last day covered by the cached harvest. Harvest and screening were completed on the 6 October run; this run authored from that cache without re-querying. **Dailies for 7 and 8 October have not yet been harvested** and the W40 weekly (27 September–3 October) is still owed; both follow.

Corpus analytics

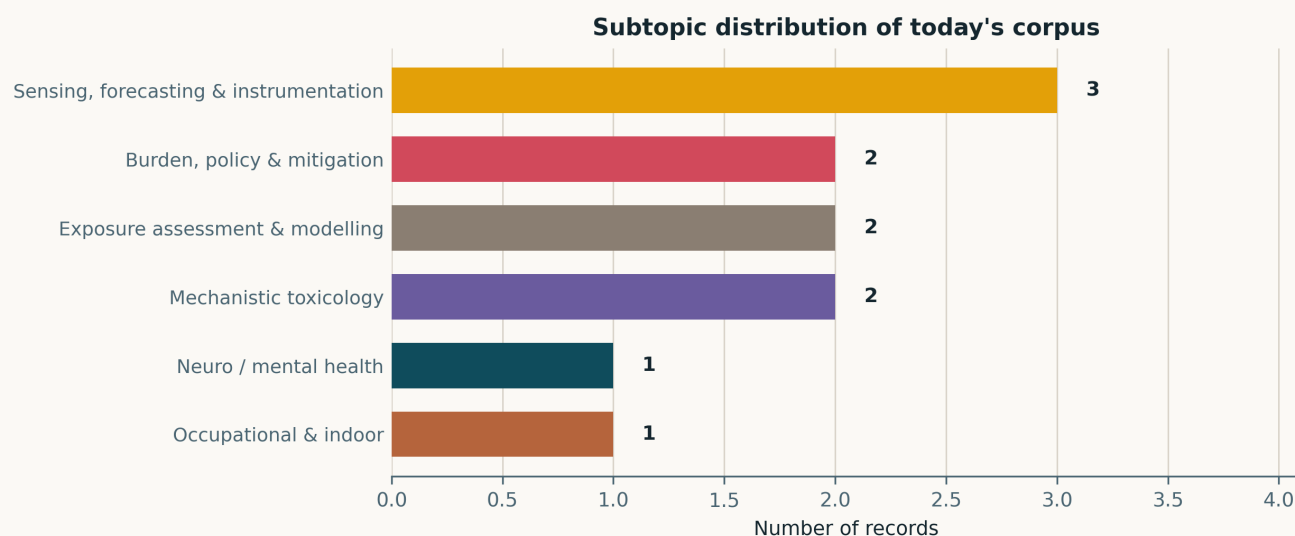


Fig. Subtopic assignment of the 11 in-scope records. **Six of ten bands populated**; *Sensing* leads with three, *Burden*, *Exposure assessment* and *Mechanistic toxicology* two each, and one each for *Neuro* and *Occupational & indoor*. The measurement side holds 8 of 11 records — the highest share of the backfill.

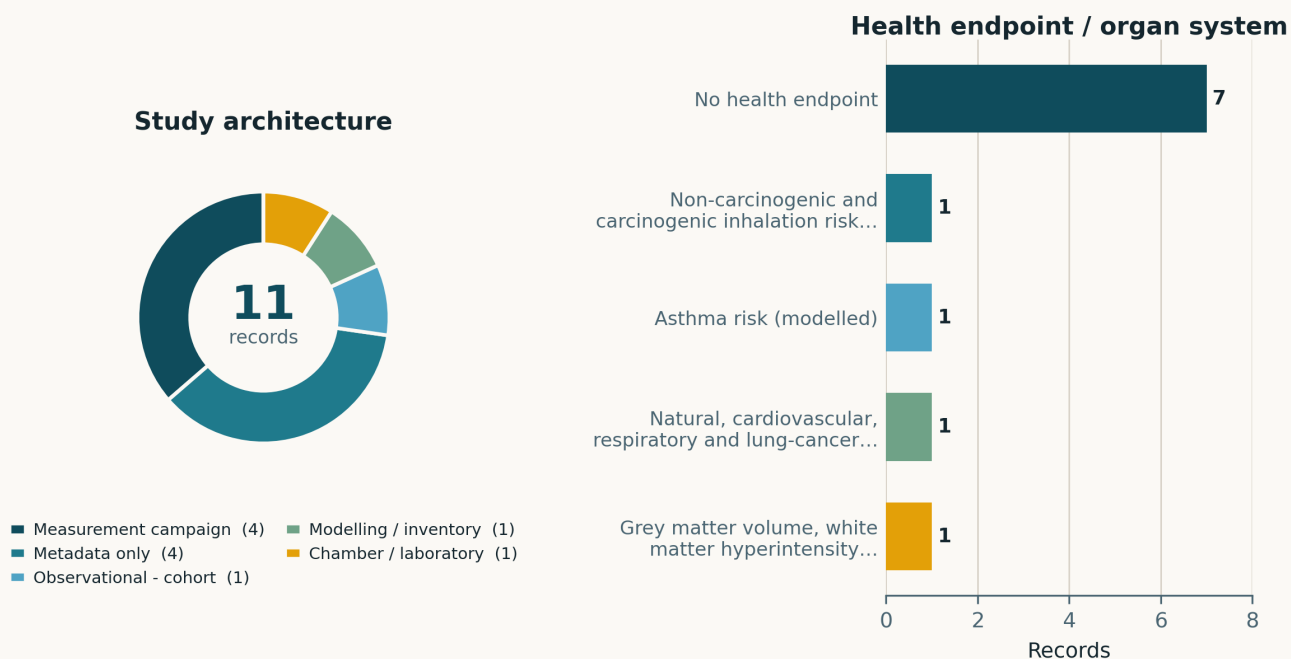


Fig. Left — architecture: **four** measurement campaigns (the AMS receptor site, the single-particle series, the school indoor-outdoor survey and the Kazakh elemental sampling), **four** metadata-only deposits, and one each for the imaging cohort, the national health-impact model and the coal-fire laboratory rig. Right — only two records report a health endpoint, and one of those (the brain-imaging cohort) reports it without a number.

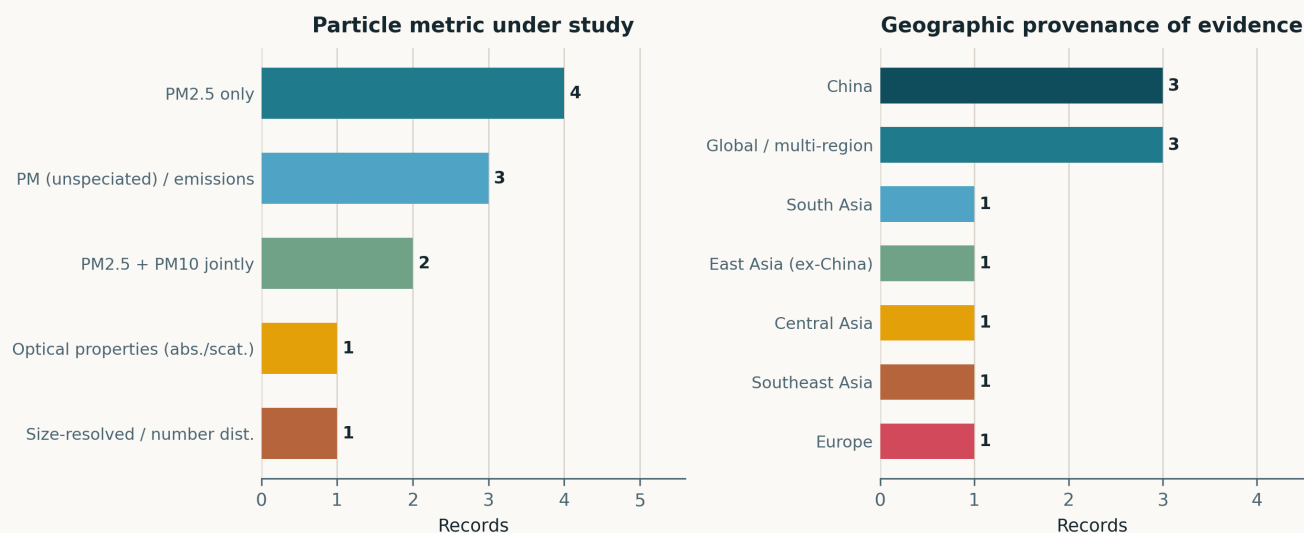


Fig. Left — particle metric: PM_{2.5} only **four**, unspicated PM or emissions three, PM_{2.5} + PM₁₀ jointly two, and one each for size-resolved number distribution (the submicron AMS record) and optical properties (the HULIS deposit). Right — geography: China and global / not stated **three** each, and one each for Europe, Southeast Asia, Central Asia, East Asia excluding China, and South Asia — Kazakhstan is the first Central Asian record since 29 September.

Life-course windows addressed today (records may span several stages)

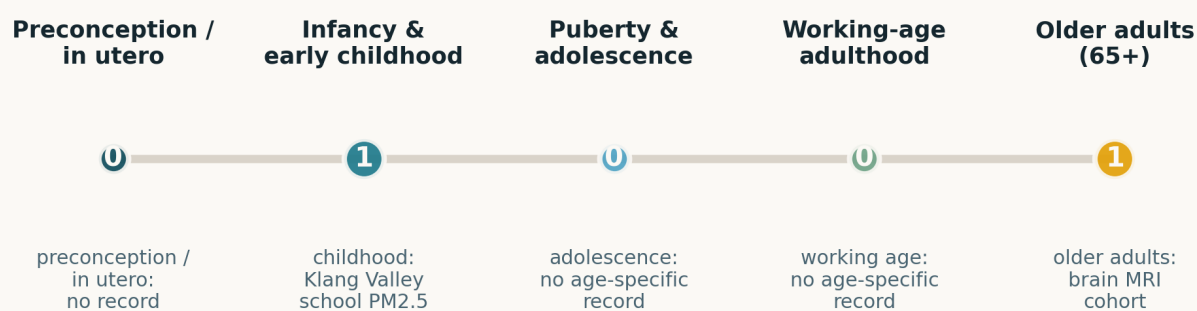


Fig. Life-course windows: childhood one (the Klang Valley school measurement) and older adults one (the repeated-MRI cohort). In utero, adolescence and working age have no record — a day dominated by atmospheric measurement rather than by population studies.

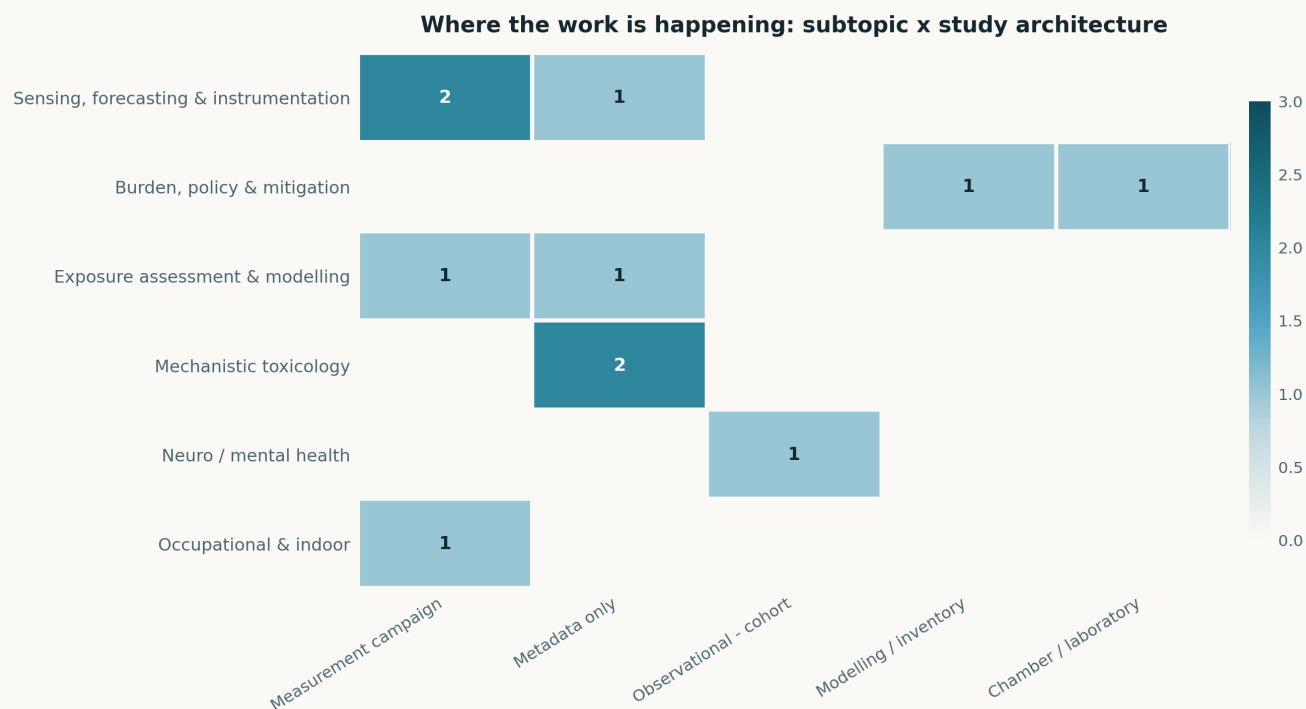


Fig. Subtopic × study architecture for the 11 records. Measurement campaigns occupy sensing, exposure and occupational; the metadata-only block spans sensing, exposure and mechanistic toxicology; burden holds the only two modelling-type records, one national and one at bench scale.

No forest plot this issue. No admitted abstract reports a PM effect estimate with a confidence interval. The brain-imaging cohort is the only record with an epidemiological estimand, and its abstract reports direction without magnitude, interval or exposure contrast; the Kazakh hazard quotients and the Italian attributable fractions are risk-assessment outputs, not estimated associations, and are excluded from the panel by the same rule that has excluded them all series. The f5 panel is absent by design, not by build failure.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health

1 record

Wang et al. 2026c Environ Pollut

Long-Term Air Pollution Exposure and Brain Structural Decline: A Longitudinal Analysis

4,276 middle-aged and older adults with repeated brain MRI; global, grey- and white-matter volumes and white-matter hyperintensity burden regressed on modelled residential PM_{2.5} and PM₁₀ in fully adjusted multivariable models. Higher long-term particulate exposure was associated with accelerated global and regional grey-matter loss and greater WMH progression, which the authors read as both a neurodegenerative and a cerebrovascular pathway. Limitation: the abstract gives no effect sizes, no confidence intervals, no exposure contrast and does not name the cohort or country - unusual for a quantitative imaging analysis, and it makes the result unassessable as reported; repeat-MRI subsamples are also selected for survival and compliance. Relevance: moderate - brain structure is the mediator between the UFP dementia-mortality signal and clinical disease, but this record cannot be entered as an effect estimate.

doi: 10.1016/j.envpol.2026.129269

Mechanistic toxicology

2 records

Sharma et al. 2026 Aerosol Sci Technol

Characterization of carbonaceous particles generated from three flame modes using a single burner to enable properties-based exposure studies

METADATA ONLY. Metadata-only record: Taylor & Francis deposited title, authors and DOI on 6 Oct with no abstract. By title, a single burner operated in three flame modes to generate carbonaceous particles of controlled and differing properties, explicitly to support properties-based rather than mass-based exposure studies. That framing is the open problem behind every toxicology record in this backfill: dose expressed as mass hides composition, and a generator that varies one property at a time is how it gets separated. Relevance: high; on the retry list.

doi: 10.1080/02786826.2026.2737177

Yadav et al. 2026 Environ Pollut

The dual role of atmospheric HULIS in inducing climate forcing and oxidative stress in the northwestern Himalayas

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 6 Oct with no abstract. By title, HULIS measured for both light absorption (brown-carbon forcing) and oxidative stress, in a high-altitude Himalayan setting. HULIS is the fraction that links the optical property a low-cost sensor responds to with the oxidative potential that toxicology implicates, so a single study reporting both is more useful than two reporting one each. Relevance: high; on the retry list.

doi: 10.1016/j.envpol.2026.129300

Exposure assessment & modelling

2 records

Mukhamediyarov et al. 2026 Atmosphere*Elemental Composition, Sources, and Health Risks of Ambient PM_{2.5} in Ust-Kamenogorsk (Oskemen) and Pavlodar, Kazakhstan*

PCA separates a non-ferrous metallurgical profile in Ust-Kamenogorsk from coal fly ash, ferrous metallurgy and petrochemicals in Pavlodar. The risk arithmetic then reports cumulative hazard indices exceeding 1 by more than 500-fold, driven by Zn (HQ_{inh} 304.7) and Al (129.5), and carcinogenic risks for Cr and As up to 9x the 1e-4 threshold. Limitation: a hazard quotient of 300 for zinc is a sign that the reference concentration is being misapplied, not that residents are inhaling a toxic zinc dose - EPA inhalation RfCs do not exist for several of these elements and the substitutes used are oral values scaled by default breathing rates; sampling duration and n are not stated. Relevance: moderate for the Central Asian composition data, which is genuinely scarce; the risk numbers should not be propagated.

doi: 10.3390/atmos17100977

Wang et al. 2026d Atmos Environ*Physics-inspired inductive spatiotemporal kriging for PM_{2.5} with satellite gradient constraints*

METADATA ONLY. Metadata-only record: Elsevier deposited title, authors and DOI on 6 Oct with no abstract. By title, this is the central methodological problem of a sparse network - interpolating to unmonitored locations - attacked inductively (so that new sensors can be added without retraining) and constrained by satellite gradients rather than by a stationarity assumption. The Everglades record in the 2 October issue measured exactly how badly ordinary kriging and IDW fail day to day; this is the proposed repair. Relevance: high, the highest of any metadata-only record in this backfill; flagged for a full-text read as soon as the abstract appears.

doi: 10.1016/j.atmosenv.2026.122398

Sensing, forecasting & instrumentation

3 records

Loh et al. 2026 Mar Pollut Bull*Seasonal characteristics of submicron aerosols in the coastal atmosphere of southeastern Korea: Influence of urban-port outflow*

Seasonal HR-ToF-AMS measurement at a coastal receptor site southwest of Busan, winter 2021 to autumn 2022. PM₁ ranged 4.23-10.0 $\mu\text{g m}^{-3}$; PMF resolved five organic factors including hydrocarbon-like, oxidised primary (OPOA), two oxygenated and biomass-burning. Ultrafine growth events were frequent, most in autumn, when OPOA reached 36% alongside elevated black carbon under north-easterly transport from the port. Limitation: one receptor site with no upwind reference, so 'port influence' rests on wind sector rather than on a source-resolved tracer, and AMS does not measure refractory sea salt or dust - the submicron mass is non-refractory by construction. Relevance: high - growth events at 4-10 $\mu\text{g m}^{-3}$ of mass are invisible to a mass-reporting network, which is the recurring theme of this week's issues.

doi: 10.1016/j.marpolbul.2026.120437

Dai et al. 2026 Atmos Chem Phys*Cross-phase partitioning of sulfur-nitrogen ratios and aerosol mixing-state evolution based on single-particle observations*

SPA-MS through winter and summer emission-control periods and matched normal periods, with three sulfur-to-nitrogen metrics (gas gSNR, particle pSNR, number-based nSNR) analysed by causal inference plus interpretable machine learning. Control periods cut NO_x while SO₂ stayed roughly flat, raising gSNR, lifting pSNR by about 57% across particle classes and expanding mean vacuum aerodynamic diameter by 27%; enrichment was greatest in particles containing both black and organic carbon and least in BC-free inorganic particles. Limitation: emission-control periods are confounded with season and meteorology, and 'causal inference' on an observational time series with co-varying controls is a strong label for what remains an association. Relevance: high - a 27% diameter shift under a control policy is a direct change in the optical cross-section a nephelometric sensor measures, i.e. a calibration drift caused by policy rather than by hardware.

doi: 10.5194/acp-26-14015-2026

Zhang et al. 2026e Appl Opt*Full field of view temporal calibration for the directional polarimetric camera onboard the GaoFen-5 (02) satellite based on a data reconstruction method*

METADATA ONLY. Metadata-only record: PubMed indexed title, authors and DOI on 6 Oct; Applied Optics deposited no abstract and the retry leg recovered none. By title, a full-field-of-view temporal calibration for the DPC instrument, correcting on-orbit response drift by data reconstruction. Polarimetric retrievals are the satellite side of aerosol loading, and a drifting on-orbit calibration propagates straight into the AOD that surface PM_{2.5} models are trained against - the same class of problem as field drift in a ground sensor, at a different scale. Relevance: moderate-high; on the retry list.

doi: 10.1364/ao.597069

Occupational & indoor

1 record

Othman et al. 2026 Indoor Air*Impact of Urban Land Use on Indoor Air Quality: Quantifying Ambient Particle Infiltration and Asthma Risk in Naturally Ventilated School Buildings*

Ten primary schools stratified by land use (high-traffic, industrial, residential, mixed industrial/high-traffic, mixed industrial/port) with paired indoor-outdoor PM_{2.5} and PM₁. Indoor maxima were at mixed industrial/port schools (24 h PM_{2.5} 35.33 +/- 9.25, PM₁ 23.79 +/- 5.87 $\mu\text{g m}^{-3}$); outdoor maxima at high-traffic schools (34.50 +/- 20.64 and 23.78 +/- 13.09); land-use class differences were significant (p<0.05) and the mean infiltration factor exceeded 0.6. Limitation: ten buildings across five classes is two per class, so the land-use contrast is confounded with building; Finf estimated from indoor-outdoor ratios without an independent air-change measurement absorbs any indoor source into the infiltration term. Relevance: high - Finf > 0.6 in naturally ventilated classrooms means outdoor network data is a usable proxy for children's exposure in this climate, which is the assumption most school studies make without testing.

doi: 10.1155/ina/5050009

Burden, policy & mitigation

2 records

Ranzi et al. 2026 Epidemiol Prev*Atlante Aria e Salute: methods for estimating the health impacts of air pollution and national application*

Methodological description of the Air and Health Atlas, an interactive national platform. PM_{2.5} and NO₂ are modelled at 1 km with random forests, converted to population-weighted exposure at municipality level against 2021 census blocks, and combined with provincial death-certificate rates through HRAPIE-2 concentration-response functions against the 5 µg m⁻³ WHO counterfactual; 2019 is shown as an illustration, with pronounced north-south gradients. Limitation: population-weighted exposure at municipality scale cannot capture within-municipality contrast, the random-forest surface is validated against the same monitors that train it, and HRAPIE-2 functions are transferred from other populations. Relevance: high - this is the operational template a low-cost network would be asked to improve, and the 1 km population-weighted step is exactly where added density pays.

doi: [10.19191/ep26.4-5.s1.a1030.092](https://doi.org/10.19191/ep26.4-5.s1.a1030.092)**Zhao et al. 2026e** Research Square (preprint)*Thermal-oxygen evolution and atmospheric pollutant emission characteristics during the vertical propagation of underground coal fires*

A scaled physical-similarity rig run from ignition to burnout with continuous monitoring at six depths. Thermal-oxygen evolution split into incubation, rapid heating, intensive reaction, rapid cooling and burnout; the upper layers burned out while the middle sustained reaction and the deepest was oxygen-limited; surface response lagged the seam. Particulate matter and light hydrocarbons were released early (CH₄ and C₂H₆ concentrated in the first 40-60 h, at background by ~80 h) while CO₂ and CO dominated later. Limitation: a similarity-scaled rig reproduces sequence, not magnitude - emission factors from it cannot be transferred to a real coal fire, and no scaling validation is offered; preprint. Relevance: moderate - the early-PM, late-CO sequence is a usable detection signature for surface sensor arrays over suspected underground fires.

doi: [10.21203/rs.3.rs-10517753/v1](https://doi.org/10.21203/rs.3.rs-10517753/v1)

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Wang et al.	Long-Term Air Pollution Exposure and Brain Structural Decline: A Longitudinal Analysis	<i>Environ Pollut</i>	Prospective cohort	Residential PM _{2.5} and PM ₁₀ from high-resolution spatiotemporal models	Not stated in abstract (middle-aged and older adult imaging cohort)
Mechanistic toxicology					
Sharma et al.	Characterization of carbonaceous particles generated from three flame modes using a single burner to enable properties-based exposure studies	<i>Aerosol Sci Technol</i>	Metadata only (no abstract)	Metadata only - carbonaceous particles from three flame modes on one burner (by title)	Not recoverable from metadata
Yadav et al.	The dual role of atmospheric HULIS in inducing climate forcing and oxidative stress in the northwestern Himalayas	<i>Environ Pollut</i>	Metadata only (no abstract)	Metadata only - humic-like substances, light absorption and oxidative potential (by title)	India (north-western Himalayas)
Occupational & indoor					
Othman et al.	Impact of Urban Land Use on Indoor Air Quality: Quantifying Ambient Particle Infiltration and Asthma Risk in Naturally Ventilated School Buildings	<i>Indoor Air</i>	Field measurement	Paired indoor-outdoor 24 h PM _{2.5} and PM ₁ with infiltration factor, 10 schools across five land-use classes	Malaysia (Klang Valley)
Sensing, forecasting & instrumentation					
Dai et al.	Cross-phase partitioning of sulfur-nitrogen ratios and aerosol mixing-state evolution based on single-particle observations	<i>Atmos Chem Phys</i>	Long-term field measurement campaign	Single-particle aerosol mass spectrometry with gas-, particle- and number-based S/N ratios	China (Yangzhou, eastern China)
Loh et al.	Seasonal characteristics of submicron aerosols in the coastal atmosphere of southeastern Korea: Influence of urban-port outflow	<i>Mar Pollut Bull</i>	Long-term field measurement campaign	PM ₁ mass, composition and size distribution by HR-ToF-AMS, four seasons, with PMF	South Korea (Geoje Island, downwind of Busan)
Zhang et al.	Full field of view temporal calibration for the directional polarimetric camera onboard the GaoFen-5 (02) satellite based on a data reconstruction method	<i>Appl Opt</i>	Metadata only (no abstract)	Metadata only - on-orbit temporal calibration of a directional polarimetric camera (by title)	China (GaoFen-5 (02) mission)
Exposure assessment & modelling					
Mukhamediyarov et al.	Elemental Composition, Sources, and Health Risks of Ambient PM _{2.5} in Ust-Kamenogorsk (Oskemen) and Pavlodar, Kazakhstan	<i>Atmosphere</i>	Source apportionment	PM _{2.5} elemental composition by ICP-OES and ICP-MS with PCA source resolution	Kazakhstan (Ust-Kamenogorsk, Pavlodar)
Wang et al.	Physics-inspired inductive spatiotemporal kriging for PM _{2.5} with satellite gradient constraints	<i>Atmos Environ</i>	Metadata only (no abstract)	Metadata only - inductive spatiotemporal kriging for PM _{2.5} under satellite gradient constraints (by title)	Not recoverable from metadata
Burden, policy & mitigation					
Ranzi et al.	Atlante Aria e Salute: methods for estimating the health impacts of air pollution and national application	<i>Epidemiol Prev</i>	Health impact model	Modelled annual PM _{2.5} and NO ₂ at 1x1 km, random forest over monitors, satellite and covariates, 2016-2024	Italy (national, municipality level)

First author	Focus	Journal	Design	Metric	Region
Zhao et al.	Thermal-oxygen evolution and atmospheric pollutant emission characteristics during the vertical propagation of underground coal fires	Research Square (preprint)	Chamber / laboratory	PM _{2.5} , PM ₁₀ and TSP with CO ₂ , CO, CH ₄ and C ₂ H ₆ through a simulated burn cycle	China (laboratory, physical similarity simulation)

Provenance and method

Entry window **6 October 2026**, single day, continuous with the 5 October issue. Harvest and screening were completed on the **6 October** run and cached (`cache/2026-10-06/_fresh.json`); this run authored from that cache without re-querying. **72** fresh candidates after cross-day deduplication, including **7** appended from the PubMed connector; **OpenAlex** HTTP 429 again; **arXiv** answered (148 kB) with no entry dated 6 October passing the recency gate. **11 admitted, 61 rejected** with logged reasons. **Four records carried on metadata alone** (36%) — the fourth consecutive issue above 25%, and one of them (*Atmos. Environ.*, satellite-constrained inductive kriging) is the highest-relevance record of the day despite having no abstract. **Consensus** returned 10 calibration hits, **0 new**. **Scholar Gateway** fails with `ACCESS_DENIED`. **ClinicalTrials.gov** (LastUpdatePostDate 3–6 October): EPIC-AIR NCT07500948 and a CAN-EXACT update, both already in the trial watch. An *Atmos. Environ.* retraction notice (drought and PM_{2.5} in Texas) appeared in the window and concerns no record in this corpus. DOI gate: **0 fail, 1 warn** — the *Epidemiologia & Prevenzione* supplement DOI resolves but is not in Crossref, which is normal for that publisher and was confirmed by eye. Abstracts remain the copyright of their respective publishers.